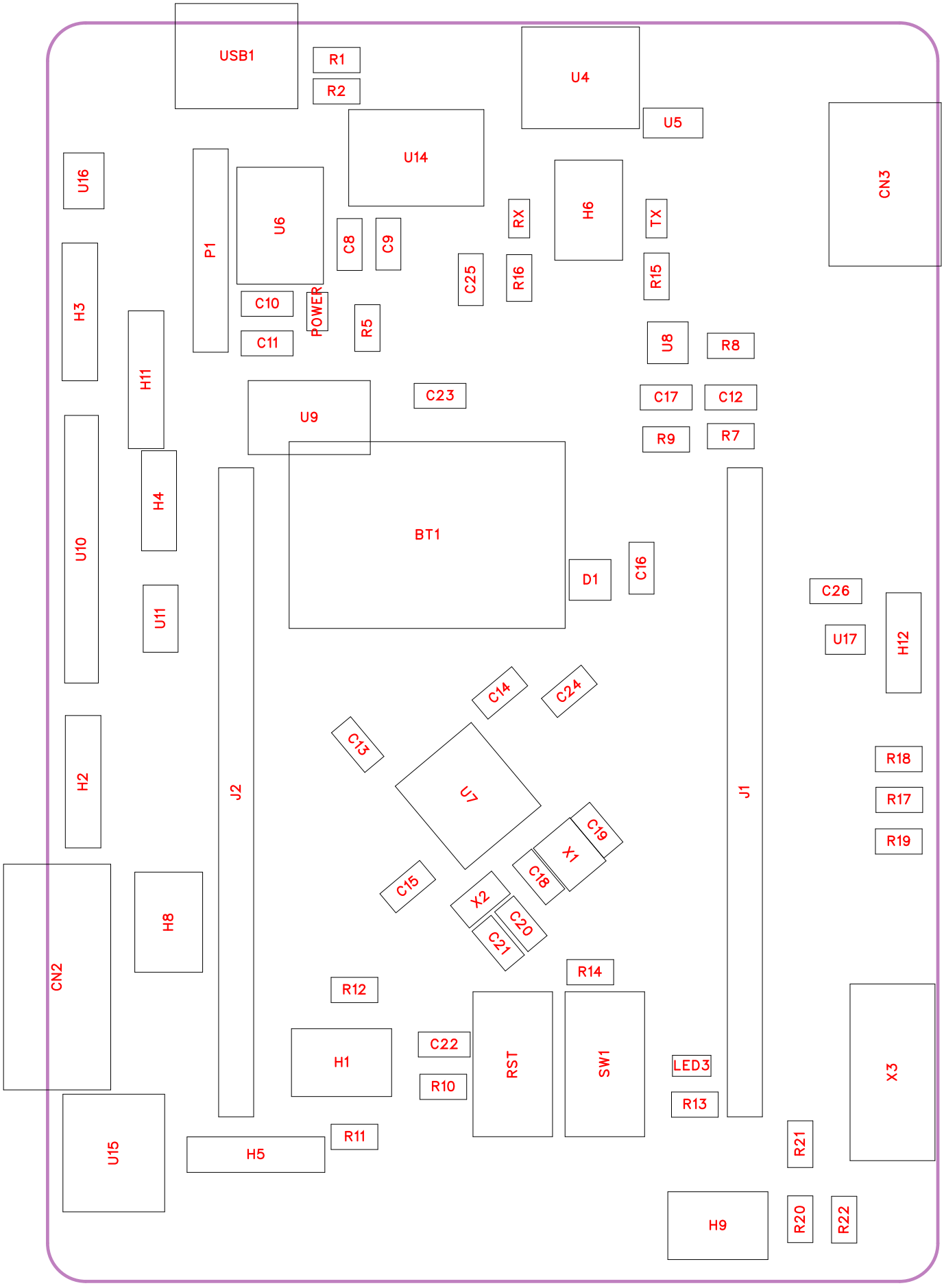
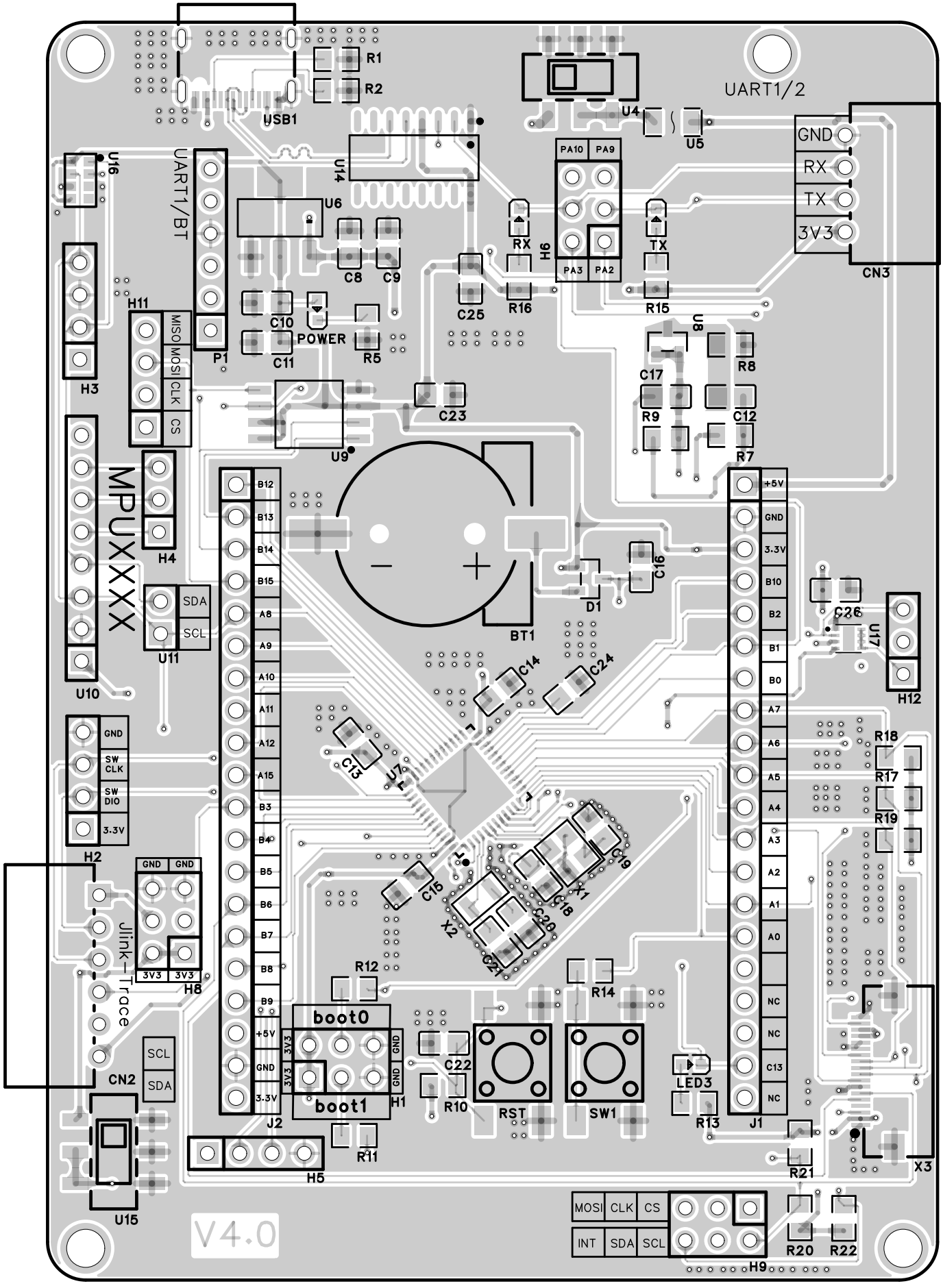
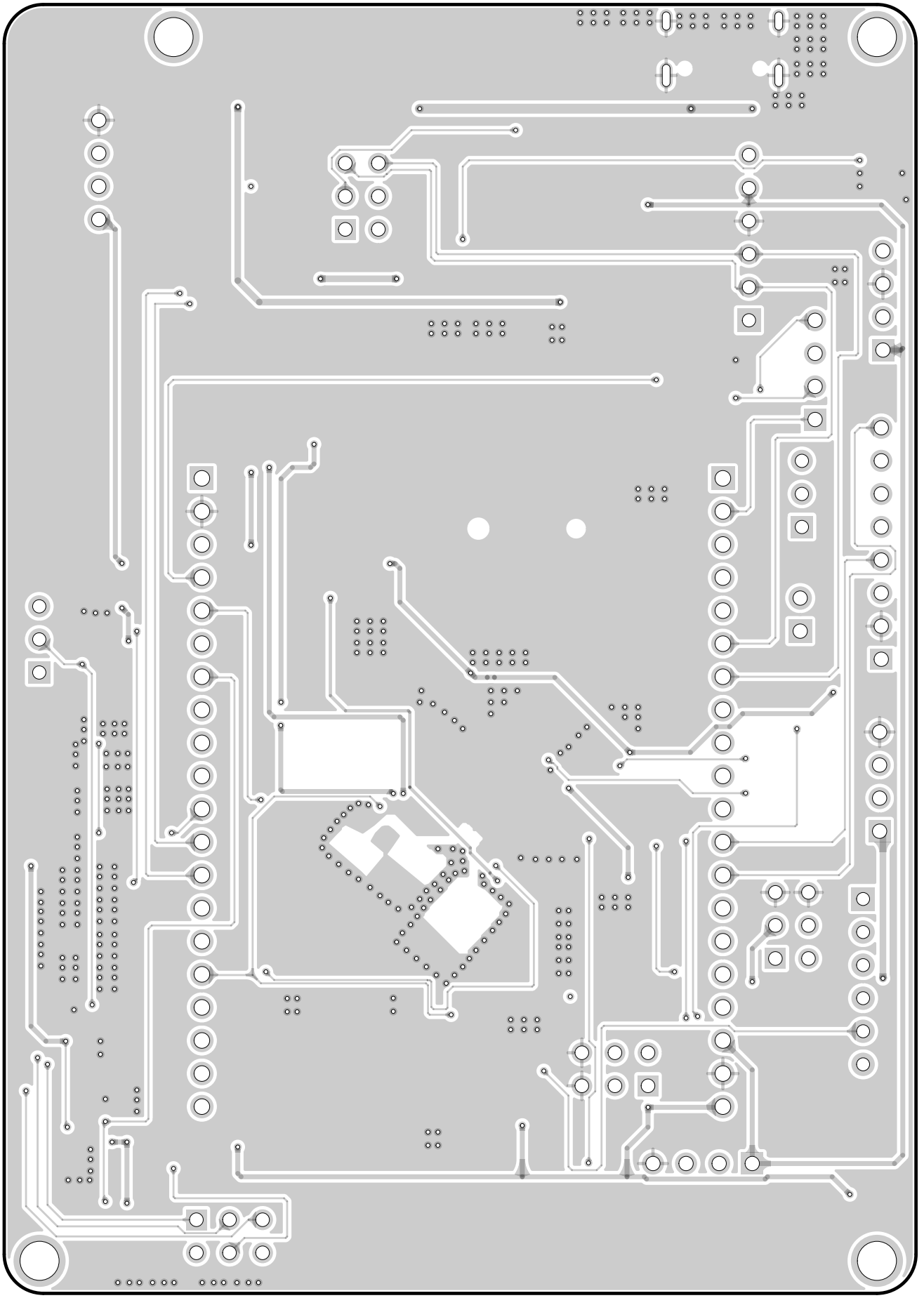
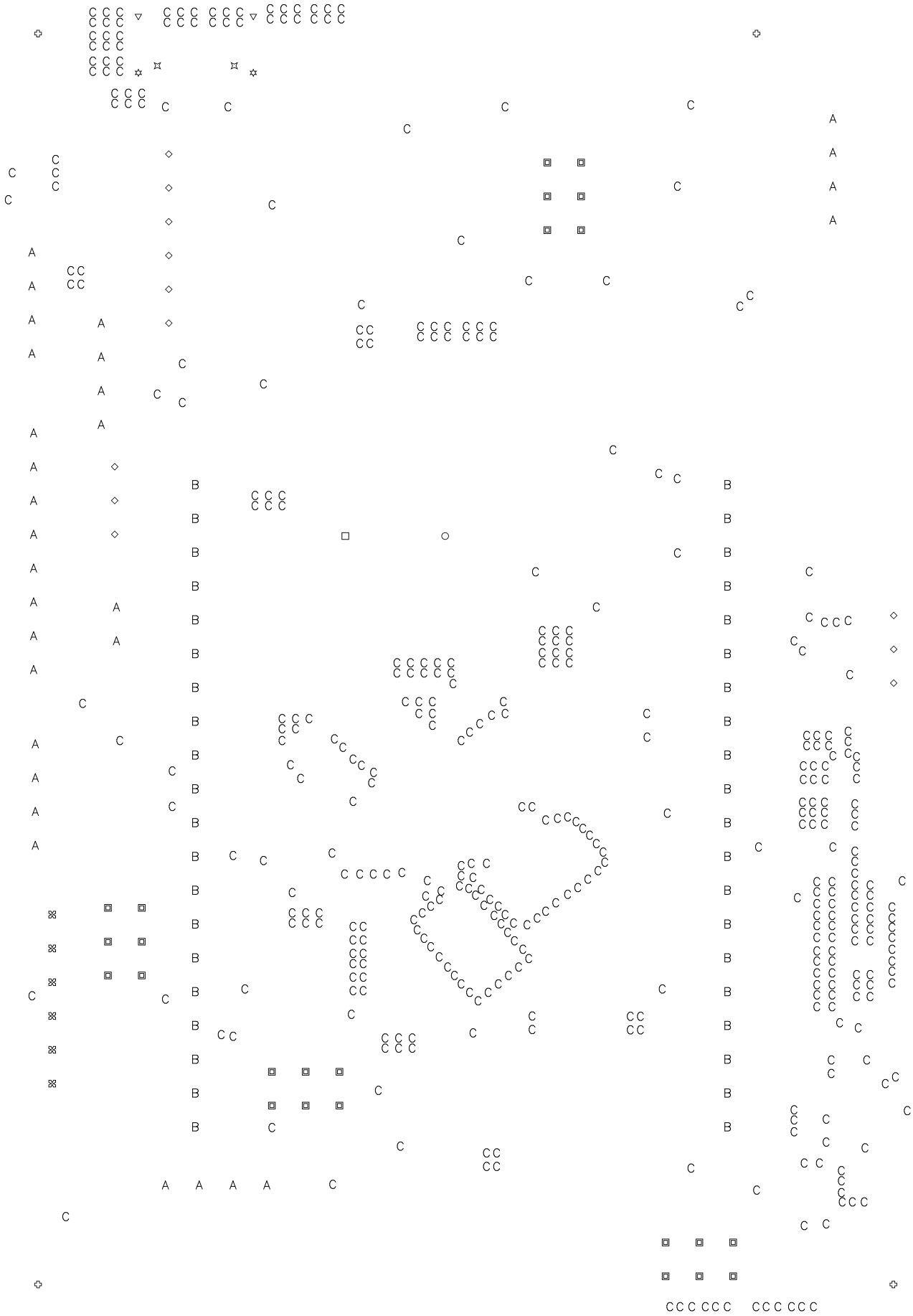

USB1
R1
R2
U4
U5
CN3
U14
U16
P1
U6
H6
TX
RX
C8
C9
C25
R16
R15
H3
C10
POWER
C11
R5
U8
R8
H11
U9
C23
C17
C12
R9
R7
U10
H4
BT1
D1
C16
C26
U11
U17
H12
C14
C24
C13
U7
R18
H2
J2
J1
R17
C19
X1
R19
C18
C15
X2
C20
C21
H8
CN2
R14
R12
H1
C22
RST
SW1
LED3
X3
R10
R13
U15
H5
R11
R21
H9
R20
R22

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	1	CR1220-2ZX	BT1	BAT-SMD_CR1220-2ZX		CR1220-2ZX	QIJEY	C969906	LCSC
2	3	10uF	C8,C10,C17	C0805	10uF				
3	10	100nF	C9,C11,C12,C13,C14,C15,C16,C22,C23,C25	C0805	100nF				
4	2	20pF	C18,C19	C0805	20pF				
5	2	10pF	C20,C21	C0805	10pF				
6	1	2.2uF	C24	C0805	2.2uF				
7	1	1uF	C26	C0805	1uF				
8	1	XH-6AW	CN2	CONN-TH_6P-P2.54_HCTL_XH-6AW_C2908615		XH-6AW	HCTL(华灿天禄)	C2908615	LCSC
9	1	XH-4AW	CN3	CONN-TH_XH-4AW		XH-4AW	HCTL(华灿天禄)	C2908613	LCSC
10	1	BAT54C	D1	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		BAT54C		C9900013520	LCSC
11	4	HDR-M_2.54_2x3	H1,H6,H8,H9	HDR-TH_6P-P2.54-V-R2-C3-S2.54_A2541WV-2X3P				C402781	
12	1	HDR-M_2.54_1x4P	H2	HDR-TH_4P-P2.54-V-M				C492403	
13	3	2.54-1*4P	H3,H5,H11	HDR-TH_4P-P2.54-V-F		2.54-1*4P	BOOMELE(博穆精密)	C2718488	LCSC
14	2	PZ254V-11-03P	H4,H12	HDR-TH_3P-P2.54-V-M_PZ254V-11-03P		PZ254V-11-03P	XFCN(兴飞)	C2937625	LCSC
15	2	2.54-1*20P直针	J1,J2	HDR-TH_20P-P2.54-V-M-1		2.54-1*20P直针	BOOMELE(博穆精密)	C50981	LCSC
16	1	LED_0603-B	LED3	LED_0603					
17	1	2.54-1*6P	P1	HDR-TH_6P-P2.54-V-F		2.54-1*6P	BOOMELE(博穆精密)	C40877	LCSC
18	3	LED_0603-G	POWER,RX,TX	LED_0603					
19	2	5.1K	R1,R2	R0805	5.1K				
20	1	2K	R5	R0805	2K				
21	2	0	R7,R8	R0805	0				
22	3	1K	R9,R15,R16	R0805	1K				
23	9	10K	R10,R11,R12,R17,R18,R19,R20,R21,R22	R0805	10K				
24	1	1.5K	R13	R0805	1.5K				
25	1	330	R14	R0805	330				

26	2	Key_SMD_6x6x6.5	RST,SW1	SW-SMD_4P-L6.0-W6.0-P4.50-LS9.0-2					
27	2	YSSP2AG00	U4,U15	SW-SMD_YSSP2AG00		YSSP2AG00	YIYUAN(怡远)	C5188558	LCSC
28	1	JK-nSMD005/60V	U5	F1206		JK-nSMD005/60V	JK(金科)	C2830235	LCSC
29	1	AMS1117-3.3	U6	SOT-223-3_L6.5-W3.4-P2.30-LS7.0-BR		AMS1117-3.3	AMS	C6186	LCSC
30	1	STM32F411CEU6	U7	UFQFPN-48_L7.0-W7.0-P0.50-BL-EP		STM32F411CEU6	ST(意法半导体)	C60420	LCSC
31	1	TL431	U8	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		TL431	YFSEMI(江苏佑风微)	C2887585	LCSC
32	1	W25Q64JVSSIQSOIC8	U9	SOIC-8_L5.3-W5.3-P1.27-LS7.9-BL		W25Q64JVSSIQSOIC8		C9900009206	LCSC
33	1	2.54-1*8P	U10	HDR-TH_8P-P2.54-V-F		2.54-1*8P	HCTL(华灿天禄)	C2897371	LCSC
34	1	HX PZ2.54-1x2PZZ	U11	HDR-TH_2P-P2.54-V-M-3		HX PZ2.54-1x2PZZ	hanxia(韩下)	C32713268	LCSC
35	1	CH340C	U14	SOP-16_L10.0-W3.9-P1.27-LS6.0-BL		CH340C	WCH(南京沁恒)	C84681	LCSC
36	1	EM7028	U16	DFN-8_L4.2-W1.4-P0.95-TL		EM7028	艾普柯	C91478	LCSC
37	1	SGM3132YDE8G/TR	U17	TDFN-8_L2.0-W2.0-P0.50-BL-EP		SGM3132YDE8G/TR	SGMICRO(圣邦微)	C5141367	LCSC
38	1	HC-TYPE-C-16P-01B	USB1	USB-C-SMD_KH-TYPE-C-16P		HC-TYPE-C-16P-01B	HCTL(华灿天禄)	C2894898	LCSC
39	1	25MHz	X1	CRYSTAL-SMD_4P-L3.2-W2.5-BL	25MHz	X322525MOB4SI	YXC(扬兴晶振)	C9006	LCSC
40	1	32.768kHz	X2	OSC-SMD_L3.2-W1.5	32.768kHz	SF32WK32768D31T002	TKD(泰晶)	C271654	LCSC
41	1	AFC01-S18FCA-00	X3	FPC-SMD_18P-P0.50_JS_AFC01-S18FCA-00		AFC01-S18FCA-00	JS(钜硕电子)	C262666	LCSC


R1
R2
USB1
U4
U5
UART1/2
GND
RX
TX
3V3
CN3
U16
UART1/BT
U14
U6
PA10
PA9
PA3
PA2
H6
RX
TX
C8
C9
C25
R16
R15
C10
POWER
R5
C11
P1
H11
MISO
MOSI
CLK
CS
H3
U8
C17
R8
R9
C12
C23
U9
R7
MPUXXXX
H4
B12
B13
B14
B15
A8
A9
A10
A11
A12
A15
B3
B4
B5
B6
B7
B8
B9
+5V
GND
3.3V
BT1
C16
D1
+5V
GND
3.3V
B10
B2
B1
B0
A7
A6
A5
A4
A3
A2
A1
A0
NC
NC
C13
NC
C26
U17
H12
SDA
SCL
U11
U10
C14
C24
C13
U7
R18
R17
R19
GND
SW CLK
SW DIO
3.3V
H2
GND
GND
3V3
3V3
H8
Jlink-Trace
C15
X1
C18
C19
X2
C20
C21
R12
R14
boot0
3V3
3V3
GND
GND
boot1
H1
C22
R10
RST
SW1
LED3
R13
J1
J2
R11
H5
R21
X3
SCL
SDA
CN2
U15
V4.0
MOSI
CLK
CS
INT
SDA
SCL
H9
R20
R22